

Digital Egypt Youth (DEY) HireReady Initiative

RAG Knowledge Base

Program Guidelines & Track Catalogue — National Telecommunication Institute (NTI)

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1. About the Initiative

The Digital Egypt Youth (DEY) initiative plays a vital role in achieving one of the strategic objectives of the ICT sector by empowering young Egyptians to thrive in the digital economy, contributing to the country's socioeconomic development and digital transformation initiatives.

The Digital Egypt Youth (DEY) initiative aims to provide young Egyptians with the necessary skills and knowledge to thrive in the digital age. The initiative focuses on training youth in various digital fields to increase their employability and entrepreneurship opportunities. It specifically targets young Egyptians, equipping them with in-demand skills needed for jobs in the digital economy, and aims to bridge the digital skills gap in Egypt to create a digitally skilled workforce supporting the country's digital transformation.

The initiative offers a full-fledged scholarship and targets 2,500 fresh graduates from Egyptian universities nationwide. The initiative adopts an upskilling approach covering two aspects: technical upskilling and personal/soft-skills upskilling, seeking to cover various digital skills to cater to diverse interests and career paths.

DEY collaborates with industry partners and experts to develop and deliver relevant training content, ensuring the program aligns with industry requirements and equips participants with practical, in-demand skills. The program facilitates job placement by partnering with industry stakeholders, creating linkages between trained individuals and potential employers through one month of on-the-job training at the industry partner's premises.

NTI has collaborated with more than 75 national and multinational industry partners — including Amazon (AWS), Etisalat Egypt, Vodafone, Huawei, WE, Cisco, VMware, Orange, Valeo, Avelabs, SEITech, Brightskies, and VOIS, among others — to design more than 30 professional tracks that meet job profiles and ICT market needs. At the end of the program, participants receive joint certification from NTI and the industry partners, validating their skills and competencies for future job opportunities.

A key target field of the Digital Egypt Youth initiative is Electronics & Embedded Systems, matching the presidential HireReady initiative.

Key facts:

- Full-fledged scholarship for selected fresh Egyptian university graduates, within **five calendar years** of graduation.
- Collaboration with **75+** national and multinational industry partners.
- More than **30 professional tracks**, each designed with industry partners to meet job profiles and ICT market needs.
- Each track = **4 months** total: Technical Program + Career Development (Soft Skills) Program, **plus 1 month of on-the-job training** at an industry partner's premises.
- Candidates join after a fine screening process conducted in cooperation with industry partners.
- Successful candidates receive **joint certificates from NTI and the industry partners**.

1.1 Categories Offered

- Electronics and Embedded Systems
- Network Infrastructure and Enablers
- Cybersecurity
- DevOps Engineering
- Telecom Engineering
- Software Engineering

1.2 Industry Partners (Selected)

WE, Orange, Vodafone, Etisalat, AWS (Amazon Web Services), Appraid, VMware, Huawei, Cisco, IBM, Brightskies, Valeo, Dell EMC, Global Brands Group, ZTE, Valleysoft, Avelabs, Sprints, RAYA, Fortinet, Aura-Tech, FEDIS, IT Visionary, Engineering Systems, Intraconsult Telecom, SecuriCIP, NEXT Career Development Academy, HITS, El Harefa, Atos, CyberX, Alfa, Vector, Sophos, Golden Technology, Techip, Intercom, SEITech, OpenValley, BTC Networks, iBravura, ConServ, Summit, Digitize, Rawafed Tech, Mentrionix, Alfostakel Technology, Visioneering, CyShield, UC Solutions, Bargasy Motors, El Delta for Electronics, Inspire Hub, iVolve, Zero&One, STS Summit

Technical Solutions, Zerosplit, Tech-Hub, geidea, Swift Act, Tejas Networks, Albetity, Kayan-IT, bmb, Koinonia, Click ITS, Codinger, Inovasy, EC-Council, Career180, BARQ Systems, Splunk, Hengtong Group, Digital Fortress, and others.

2. Common Program Structure

Every HireReady track follows the same two-part structure:

- **Part 1 - Technical Program:** 294 hours, track-specific (details per track below).
- **Part 2 - Career Development Program:** 126 hours, identical across all tracks (detailed in Section 3).

Total duration per track: **420 training hours (≈ 4 months)**, followed by **1 month of on-the-job training** at an industry partner.

3. Career Development Program (126 Hours) — Common to All Tracks

Overview

This comprehensive course is designed for aspiring professionals in the ICT field who aim to excel in international business environments. Participants develop proficiency in Business English, enabling them to confidently communicate and present in professional settings. Through hands-on simulations and workshops, attendees practice leadership and team-building skills, fostering autonomy and critical thinking.

The course also covers essential soft skills, including effective communication, time management, and problem-solving, alongside a solid foundation in project management to manage projects efficiently across industries such as Construction and Information Technology. A focus on essential workplace skills prepares participants to thrive in a corporate environment, covering teamwork, conflict resolution, and professional ethics.

3.1 Business English (35 hours)

- Speaking about Careers: discussing career paths and professional experiences.
- Maintaining a Professional Online Image: building and sustaining a positive digital presence.
- CV Writing and interview preparation with the right language tools.
- Telephoning and Arranging Appointments: phone etiquette and scheduling.
- Email Writing for various business contexts.
- Organizations and Corporations: corporate structures and functions.
- Companies' Structure and Activities: business models and operations.
- Action Plan and Proposal Writing.
- Business Meetings: conducting and participating effectively.
- Writing Meeting Minutes.

3.2 Career Development (56 hours)

Soft Skills (42 hours)

- Communication Skills (verbal and non-verbal).
- Leadership Skills and Problem Solving.
- Team Building.
- Time Management.
- Interviewing (job-interview preparation).

- Presentation Skills.
- Mentorship and Coaching.
- Emotional Intelligence.
- Stress Management.
- Workplace Legal Considerations.
- Managing Upwards.
- Fit for Work (physical/mental well-being).
- Motivation.
- Workplace Ethics.
- Change Management.
- Corporate Social Responsibility.
- Work-Life Balance.

Project Management (14 hours)

- Project Management Framework.
- Project Scope Management.
- Project Schedule Management.
- Project Cost Management.
- Project Risk Management.
- Project Monitoring and Controlling.
- Advanced Tools (PM software and methodologies).

3.3 Essential Workplace Skills (35 hours)

- **Teamwork and Collaboration:** building effective teams, role clarity and accountability, leveraging diverse skills.
- **Conflict Resolution:** identifying/addressing conflicts, mediation skills, negotiation techniques.
- **Professional Ethics:** ethical decision-making, integrity and accountability, navigating ethical dilemmas.
- **Adaptability and Flexibility:** embracing change, resilience building, creative problem solving.
- **Critical Thinking and Decision Making:** analytical skills, problem-solving strategies, decision-making models.
- **Workplace Etiquette:** professional communication, interpersonal skills, dress code and personal presentation.
- **Additional Skills for a Competitive Edge:** networking skills, advanced time management, customer service excellence, innovation and creativity.

3.4 Capstone Project (Career Development track)

Participants prepare Capstone Projects throughout the course, integrating skills and knowledge from Business English, Career Development, Project Management, and Essential Workplace Skills. On the final day (6 hours), participants present their Capstone Projects, showcasing their ability to communicate, manage projects, and demonstrate acquired skills.

4. Category: Electronics and Embedded Systems

4.1 Digital IC Design — Code: E004

Technical Program Duration: 294 Hours

Overview

Introduces students to digital electronics design and implementation on FPGAs (architecture, digital design flow, Verilog coding guidelines), with hands-on projects for basic digital blocks and complex FPGA implementations. Covers timing violations and power analysis using ISE & Vivado IDEs on Zynq-7000 development kits. Also covers ASIC design methodology (cost/size reduction through miniaturization and integration) and RTL-to-tape-out conversion, plus verification engineering: constructing a robust verification plan and complex UVM test cases for SoC designs.

Prerequisites

- Good knowledge of digital electronics design.
- Basic knowledge of HDL languages (VHDL/Verilog).
- Basic knowledge of FPGA and ASIC technology.

Contents

Module 1: Digital Design with FPGA Digital systems design review; top-down digital design flow; introduction to FPGAs and FPGA architecture; FPGA flow overview; ultrafast design methodology; VHDL/Verilog motivation, modules, operators, combinational/sequential logic; simulators; hierarchical design; blocking vs non-blocking assignments; test-benches; functions/packaging/tasks; compiler directives; system tasks; finite state machines; design constraints; timing analysis; FPGA tool design flow; Project 1 (UART); scripting languages.

Module 2: Digital Design ASIC ASIC introduction; STA basics and timing paths; FF requirements and timing constraints; EDA tool design objects; logic synthesis; formal verification; synthesis tutorial and project; floor planning, power planning, placement, ICC; CTS, routing, chip finishing; complete synthesis/formal verification/PNR/STA tutorial (2 parts); extraction and STA; EMIR and power estimation; final project.

Module 3: Digital Design Verification Verification basics; System Verilog (data types, process blocks, hierarchical structures, compiler directives, scheduling, assignments, classes, randomization, coverage); UVM overview, hierarchy, base classes, factory, phasing, resources/configurations, TLM, sequences/sequencers, components, full environment, UVM testing lab; Design for Testability (DFT).

Hands-On: Labs using FPGA kits (e.g., Zynq-7000); writing VHDL/Verilog and System Verilog programs.

4.2 Electronics Design for Industry — Code: E006

Technical Program Duration: 294 Hours

Overview

Prepares hardware designers capable of developing complete, production-ready electronic designs for smart devices (washing machines, water heaters, meters, access controls, cars, etc.). Targeted skills: design/simulation tools, sensors and actuators, linear and switched-mode power supplies, electromagnetic compatibility (EMC), PCB tools, thermal design, and reliability/lifetime estimation.

Prerequisites

- Basic knowledge of circuits.
- Basic analytical skills.
- Basic knowledge of statistics.

Contents

Basic Electronic Design (96 hr): Circuit simulation (LTSpice — transient, AC, DC, parametric, Monte Carlo); ADC types, performance, limitations; transducers and actuators; analog signal conditioning and filter design; op-amp circuits (amplifiers, active filters, level shifters, instrumentation, multi-vibrators, log, rectifiers) with practical limitations; oscillators; galvanic isolation; protection (ESD/HBM, transient protection, MOV/TVS selection); comm buses (RS232/422, I2C, SPI, UART, CAN); practical transistor circuits (BJT, MOS switches, relay drivers); heat sink sizing.

Power Supply Design (96 hr): Switching-mode power supply basics; passive/active component parasitics; non-isolated and isolated DC/DC converter design; inductor/transformer design; EMI filter design; efficiency analysis.

PCB Design for Manufacturing & EMC (96 hr): PCB design tool, component management, layer stack and layout, library/reuse; Design for Manufacturing (DFM) — manufacturing processes, reliability/lifetime calculations; Design for EMC — concepts and standards, conducted/radiated emission and immunity (setup, mechanisms, mitigation), PCB EMC design guidelines (board structure, digital/analog/high-speed/high-voltage, shielding).

Hands-On: Sensor interfacing/calibration labs; PCB design tool; 10W flyback power supply design lab; EMC labs (conducted emission evaluation); power supply lifetime estimation.

5. Category: Network Infrastructure and Enablers

5.1 Core Data Network Infrastructure — Code: N002

Technical Program Duration: 294 Hours

Overview

Trainees learn core enterprise network knowledge through hands-on experiences, starting with core services on Linux & Windows servers, then CCNP ENCOR (dual-stack IPv4/IPv6, virtualization, infrastructure, network assurance, security, automation), then CCNP ENARSI (advanced routing, Layer 3 VPN, infrastructure security/services). Security is covered via FortiGate firewall labs (FortiOS Administrator).

Targets Communication/Electronics Engineers, Computer & Information graduates, and job seekers for professional-level ICT roles. Leads to Red Hat RHCSA (EX200), Cisco 350-401 ENCOR, Cisco 300-410 ENARSI, and Fortinet NSE4 certifications.

Prerequisites

- CCNA

Contents

Microsoft Windows Server administration; CCNP Enterprise – Core Networking (ENCOR); CCNP Enterprise – Advanced Routing (ENARSI); Linux Red Hat System Administration (RHCSA); Red Hat Ansible for Network Automation; FortiOS Administrator (Fortinet NSE4).

Hands-On: Simulation tools, emulation tools, local & remote real labs.

Job Profiles: NOC Engineer, Network Engineer, System Admin Engineer, IT Infrastructure Engineer, IT Specialist Engineer.

5.2 Cloud Solutions Architect — Code: I010

Technical Program Duration: 294 Hours

Overview

Covers virtualization concepts and the journey into private/public cloud, focusing on data-center software-defined infrastructure. Targets Computer, Communications, and Electronics Engineers plus Information/Computer graduates and IT staff. Leads to Red Hat RHCSA (EX200), VMware VCP-DCV (2V0-21.23), AWS Cloud Practitioner (CLF-C01), AWS Solutions Architect Associate (SAA-C02), AWS Security Specialty (SCS-C03), Huawei HCCDA-TE and HCCDP-Cloud Migration.

Prerequisites

- Network Fundamentals (CCNA)

- Linux Foundations
- Cloud Foundations

Contents

Linux Red Hat System Administration (RHCSA); Cloud and Virtualization Concepts; VMware vSphere ICM (VCP); AWS Cloud Practitioner; AWS Solutions Architect; AWS Cloud Security; Docker & Containers & Kubernetes; Red Hat OpenShift Administration I (DO180); Huawei Cloud HCCDA-TE; Huawei Cloud HCCDP-Cloud Migration.

Hands-On: Simulation tools, virtual machines, remote labs.

Job Profiles: Cloud Engineer, Datacenter Administration Engineer, System Admin Engineer, Cloud Infrastructure Engineer, Cloud Architect Engineer.

5.3 Advanced VoIP and Collaboration Networks — Code: N004

Technical Program Duration: 294 Hours

Overview

Covers unified communications networks — voice, video, and data sessions — using Cisco and Avaya equipment. Prepares candidates for: CCNP Collaboration Call Control and Mobility (CLACCM 300-815), CCNP/CCIE Collaboration Core (CLCOR 350-801), Avaya Aura Core Components Integration (71200X), Avaya IP Office Platform Basic Integration and Configuration (77200X).

Prerequisites

- CCNA

Contents

Part I: Cisco-Based Solution AV Fundamentals (unified collaboration, SIP/H.323, RTP/RTCP/DTMF, Cisco converged collaboration); Endpoints (phones, Telepresence, registration, call settings, maintenance); Network Requirements (core network components, L2/L3 QoS, DNS/NTP/SNMP); Call Control Mechanisms (Unified CM routing/dial plans, LDAP integration, SIP registration, CAC, globalized call routing, SIP trunk config, call control features, mobility, CUBE routing/interworking, Unified CME/SRST); Edge Services (MRA, IOS XE gateway/media resources); Collaboration Applications (Unity Connection, IM & Presence, Jabber); Troubleshooting (endpoints, CUCM reports, RTMT, Disaster Recovery System, voicemail monitoring).

Part II: Avaya-Based Solution Avaya Aura overview and components; Communications Manager administration, telephones, dial plan, basic/enhanced features, COS/COR administration, group features, outgoing calls/trunks (ISDN-PRI, SIP, H.323, AAR/ARS), maintenance/security; Avaya Aura System Manager; Session Manager (SIP registration, routing policy, maintenance); Communication Manager integration (duplex config, TLS certificates); Avaya Aura Media Server; Avaya IP Office Platform Administration (manager modes, security, general/user settings, SIP trunks, short codes, button programming).

Hands-On: Exercises using Cisco Collaboration and Avaya Aura equipment.

5.4 Microsoft Azure Administration — Code: I007

Technical Program Duration: 294 Hours

Overview

Eight-course series covering core Windows Server 2016/2022 infrastructure (AD DS, networking services, Hyper-V), Azure fundamentals, Azure Administrator skills (subscriptions, VMs, storage, virtual networking, backup, hybrid connectivity, Azure AD), Azure networking design (hybrid connectivity, routing, private access, monitoring), Azure security (identity/access, platform protection, data/apps, security operations), and Windows Server Hybrid Core Infrastructure administration.

Prerequisites

- Communications and Electronics Engineering / Computer Engineering / Computer Science graduate.
- CCNA

Contents

Windows Server 2016/2022 Administration (137 hr): IPv4 network planning, DHCP, DNS, server deployment/images, local storage, domain controllers, AD DS object management, VM management, VPNs, enterprise storage, Storage Spaces/Data Deduplication, Hyper-V, advanced AD DS (sites, replication), high availability/DR, failover clustering, NLB, Group Policy, securing/monitoring/recovering AD DS.

AZ-900 Azure Fundamentals (7 hr): cloud computing/service models, Azure architecture, compute/networking/storage services, identity/access/security, cost management, governance/compliance, resource management, monitoring.

AZ-104 Azure Administrator (36 hr): Azure AD, users/groups, subscriptions, Azure Policy, RBAC, ARM templates, virtual networks/NSGs/Firewall/DNS/peering/VPN Gateway/ExpressRoute/Virtual WAN, routing/Load Balancer/App Gateway, storage accounts/Blob/security/Files, VM configuration/availability/extensions, App Services/Container Instances/AKS, backups, Monitor/alerts/Log Analytics/Network Watcher.

AZ-700 Networking Solutions (36 hr): virtual networks, hybrid networking, load balancing (HTTP/non-HTTP), network security, private access to Azure services, network monitoring.

AZ-500 Security Technologies (36 hr): Azure AD security, hybrid identity, AD identity protection, PIM, governance strategy, perimeter/network/host security, container security, Key Vault, application/storage/SQL security, Monitor, Defender for Cloud, Sentinel.

AZ-800 Hybrid Core Infrastructure (36 hr): AD DS, domain controllers/FSMO, GPOs, hybrid identity, Azure IaaS AD DCs, secure administration, JEA, hybrid workloads/Azure Arc, Hyper-V and containers, Kubernetes orchestration, IaaS VM planning/customization/automation, DHCP/DNS/IPAM/remote access, hybrid networking, file servers, Storage Spaces Direct, Data Deduplication, iSCSI, Storage Replica.

Job note: Hands-on demonstrations throughout.

5.5 NOC (Network Operations Center) Automation Engineer — Code: N006

Technical Program Duration: 294 Hours

Overview

Applies DevOps culture and automation principles to network engineering/operations (“network as API”), continuous integration/deployment of networking technologies, preparing candidates for the NOC Automation Engineer profile. Leads to Cisco 350-401 ENCOR, Cisco 300-410 ENARSI, Red Hat RHCSA (EX200), AWS Cloud Practitioner (CLF-C01), Cisco CCNP Data Center DCACI (300-620).

Prerequisites

- CCNA
- Programming Basics
- Linux Basics

Contents

AWS Academy Data Center Technician (DCT); Cloud and Virtualization Concepts; Cisco CCNP ENCOR; Cisco CCNP ENARSI; AWS Cloud Practitioner; Linux Red Hat Administration (RHCSA); Red Hat Ansible for Network Automation; Python for Network Automation; CCNP Data Center (DCACI); Network DevOps tools.

Hands-On: Simulation tools, virtual machines, remote cloud labs.

Job Profiles: Network Automation Engineer, SRE Engineer, NOC Engineer, Network Engineer, System Admin

6. Category: Cybersecurity

6.1 Defensive Security Solutions — Code: S008

Technical Program Duration: 294 Hours

Objectives

Explores cyber-criminal tactics and the technologies/products/procedures cybersecurity professionals use to combat cybercrime. Covers core network security concepts and skills for maintaining confidentiality, integrity, and availability. Extensive hands-on labs on Cisco Firepower, Fortinet FortiGate, Sophos, and Palo Alto solutions.

Prerequisites

- Engineering (Communications/Electronics or Computer Engineering), Information & Computer, IT, or Computer Science graduates.
- CCNA (certified or equivalent course).

Contents

Network and Security Fundamentals (49 hr): network design principles, IPv4/IPv6 review, router/switch config (VLANs, routing), cyber-criminal TTPs, firewall technologies, secure VPNs, IPS role, confidentiality/integrity/authentication.

Cisco Security Solution (49 hr): NGIPS/NGFW concepts; Cisco ASA (Firewall, NAT, AAA); Firepower Threat Defense setup; access control policies; security intelligence; site-to-site VPN; remote-access SSL VPN with AnyConnect; Firepower for IPS/threat management/web filtering.

Fortinet Security Solution (120 hr): - *FortiGate Administration (42 hr)*: operation modes, GUI/CLI, Security Fabric, firewall policies, NAT, user authentication, encryption/certificates, SSL/TLS inspection, application control, anti-DoS, SSL VPN, IPsec VPN, logging. - *FortiManager (21 hr)*: central management, provisioning templates, policy packages, Security Fabric integration, HA/backup/recovery, firmware management, troubleshooting. - *FortiAnalyzer (21 hr)*: central logging, architecture, access controls, HA, log management, event/incident management, reporting. - *FortiMail (28 hr)*: email security architecture, LDAP integration, SMTPS/TLS/IBE, anti-spam, anti-phishing, FortiSandbox ATP integration, DLP, dashboards. - *FortiWeb (28 hr)*: WAF fundamentals, deployment, load balancing, ML capabilities, signatures, anti-DoS/defacement, SSL/TLS, authentication, PCI DSS compliance, caching/compression, content routing/rewriting.

Sophos Security Solution (28 hr): architecture, security/NAT policies, GUI/CLI administration, network protection, site-to-site VPN, authentication, web protection/application control, remote access VPN, wireless protection, logs/reports, Sophos Central management.

Palo Alto Security Solution (28 hr): NGFW security features, security/NAT policies, certificate operations, SSL VPN, IPsec VPN, threat prevention.

Hands-On: Simulation tools, virtual machines, remote cloud labs.

Job Profiles: Network Security Engineer, Cybersecurity Engineer, Security Solutions Specialist.

6.2 Security Operation Center (SOC) Engineering — Code: S005

Technical Program Duration: 294 Hours

Overview

Teaches penetration testing and ethical hacking principles and the tools/techniques hackers use. Trains forensic analysis to trace and report evidence using Security Onion, QRadar, FortiAnalyzer, FortiSIEM, and FortiSOAR, and teaches Incident Response Plan preparation.

Prerequisites

- Engineering (Communications/Electronics or Computer Engineering), IT, or Computer Science graduates.
- CCNA (or equivalent).

Contents

Network Fundamentals (14 hr); Linux RedHat Administration-I (28 hr); FortiGate Operator (21 hr) — firewalls/NGFW, user authentication, SSL inspection, web filtering, IPS, application control, IPsec/SSL VPN; Network Penetration Testing (56 hr) — reconnaissance, scanning, vulnerability analysis, web app pentesting, social engineering, system exploitation, wireless pentesting; Cisco Cyber Security Associate (42 hr) — threat intelligence, forensic analysis, SOC operations, Security Onion, endpoint investigation, network intrusion analysis; FortiAnalyzer Analyst (21 hr); FortiSIEM (28 hr); FortiSOAR (21 hr); SPLUNK (28 hr) — searching, dashboards, correlation, reports/alerts, visualizations; IBM QRadar (35 hr) — architecture, data flows, log sources, CRE, Use Case Manager, asset management, AQL, reporting.

Hands-On: Simulation tools, virtual machines, remote cloud labs.

6.3 Red Team Penetration Tester — Code: S011

Technical Program Duration: 294 Hours

Overview

Equips participants with the penetration testing lifecycle: planning, reconnaissance, scanning, and exploitation across web applications, networks, and mobile environments. Emphasizes documentation, legal/ethical standards, and real-world problem-solving.

Target Audience / Prerequisites

- Communications and Electronics / Computer Engineering / Computer Science graduates.
- CCNA

Contents

IP Networking Review (21 hr); Linux Administration (35 hr) including Bash scripting; Python (21 hr) — data types, syntax, Jupyter, data structures, functions, classes; Firewalls and Cryptography Basics (28 hr); Web Application Fundamentals (35 hr) — HTML, CSS, client-side scripting, PHP, SQL; Web App Penetration Testing (70 hr) — authentication/access-control vulnerabilities, OS command injection, SQL injection, file upload vulnerabilities, SSRF, path traversal, XSS, CSRF, CORS; Network Penetration Testing (56 hr) — footprinting/reconnaissance, scanning, exploit/pivot, enumeration, system hacking, malware, sniffing, social engineering, DoS, evading IDS/firewalls/honeypots, wireless hacking, privilege escalation, Active Directory attacks, reporting; Mobile App Penetration Testing (28 hr) — Android pentesting.

Hands-On: Labs on virtual machines.

Job Profiles: Penetration Tester (Red Team), Vulnerability Analyst, Security Analyst, Ethical Hacker.

7. Category: DevOps Engineering

7.1 Cloud DevOps Accelerator — Code: D001

Technical Program Duration: 294 Hours

Overview

A “DevOps Accelerator” provides a proven, systematic, repeatable process to help companies reach a certain level of DevOps maturity. Leads to Red Hat RHCSA (EX200), Red Hat OpenShift Administrator (EX280), AWS Cloud Practitioner (CLF-C01), AWS Solutions Architect Associate (SAA-C03), AWS Security Specialty (SCS-C03).

Prerequisites

- CCNA
- Fundamentals of Linux and Cloud Concepts

Contents

Cloud and Virtualization Concepts; Linux Red Hat System Administration (RHCSA); AWS Cloud Practitioner; AWS Solutions Architect; AWS Cloud Security Foundations; Red Hat Enterprise Linux Automation with Ansible (RH294); Kubernetes/Helm; Red Hat OpenShift Introduction to Containers; Red Hat OpenShift Development I (DO188); Red Hat OpenShift Administration I (DO180); Infrastructure-as-Code tools (Terraform/CloudFormation/Ansible); Git; CI/CD tools (GitLab/Jenkins); Capstone Project.

Hands-On: Simulation tools, virtual machines, remote cloud labs.

Job Profiles: Cloud DevOps Engineer, DevOps/SecDevOps Engineer, System Admin Engineer, Cloud Infrastructure Engineer, OpenShift Engineer.

7.2 Professional DevSecOps Engineer — Code: D004

Technical Program Duration: 294 Hours

Overview

Embeds security throughout the software delivery lifecycle while retaining DevOps speed and agility: Linux/Bash/Python basics, Agile, Docker, Kubernetes, AWS fundamentals, Terraform, Ansible, OpenStack, and secure CI/CD. Prepares for DASA DevOps Fundamentals, Certified DevSecOps Professional, and AWS Solutions Architect Associate (SAA-C02) exams.

Target Audience / Prerequisites

- Communications and Electronics / Computer Engineering / Computer Science graduates.
- CCNA

Contents

Linux Fundamentals and Bash Scripting (35 hr) — file/user/service management, remote access, hardening; Git (7 hr); Python (22 hr) including secure coding (input validation, Paramiko, PyCrypto); Agile and DevSecOps Fundamentals (14 hr); Docker (21 hr) — secure Dockerfile creation, image scanning, secure networking, Docker Compose/Swarm, kaniko/DinD; Kubernetes (49 hr) — architecture, resource creation, networking/services, RBAC, network/pod security policies, Kubernetes security best practices, EKS, Istio service mesh/zero trust, HashiCorp Vault/AWS Secrets Manager integration; AWS Fundamentals (49 hr) — networking (VPC, subnets, routing, NAT, peering, Transit Gateway), core services (EC2/ECS/Fargate/EKS/Lambda/API Gateway/CloudFront/Route53), storage/databases, CloudFormation, monitoring, IAM security, Security Hub/GuardDuty; OpenStack Cloud Platform (21 hr); Terraform (21 hr) — IaC, state, modules, workspaces, Sentinel policies, HashiCorp Vault intro; Ansible (14 hr) — inventory, roles, hardening automation, Ansible Vault; Monitoring (7 hr) — Prometheus; CI/CD and DevSecOps Pipeline (28 hr) — Git, secure pipelines, Snyk/OWASP ZAP/Clair integration, GitLeaks, SonarQube, ArgoCD; practice project (secure end-to-end DevSecOps pipeline).

Job Profiles: DevOps/DevSecOps Engineer, AWS/OpenStack Cloud Administrator, Security-focused SRE, Cloud Security Specialist.

8. Category: Telecom Engineering

8.1 M-NOVA: Mobile Network Planning, Optimization, Virtualization and Automation — Code: W003

Technical Program Duration: 294 Hours (42 training days)

Overview

Transforms engineering graduates into enterprise-ready autonomous network engineers via academic rigor plus extensive hands-on lab work, automation, and business skills — aiming for a competence level equivalent to 2-3 years of industry experience. Core focus: end-to-end RAN Planning and Optimization integrated with Automation and AI.

Prerequisites

- Electronics and Communications Engineers.

Contents (by Phase)

Phase 1 - Deep Foundations & Legacy Networks (3G/4G LTE), Days 1-7: mobile network ecosystem, site components; 3G UMTS fundamentals and link budget; 4G LTE architecture (E-UTRAN/EPC); 4G RF planning (link budget, PCI planning); 4G optimization — accessibility/retainability, mobility management; Phase 1 milestone assessment (practical lab exam).

Phase 2 - 5G Standalone (SA) Architecture & Planning, Days 8-14: 5G SA core/RAN architecture (SBA, AMF/SMF/UPF/UDM/AUSF, CU/DU split); 5G NR physical layer and numerology; massive MIMO/beamforming; 5G radio planning and propagation modeling; 5G SA call flows/signaling; network slicing and QoS; Phase 2 milestone assessment (live lab exam).

Phase 3 - 5G SA Optimization & QoS, Days 15-21: 5G SA KPIs; coverage/capacity optimization; mobility optimization; VoNR/latency tuning; interference analysis/troubleshooting; in-building solutions/small cells; portfolio review.

Phase 4 - Network Disaggregation & O-RAN, Days 22-28: Open RAN philosophy/standards; O-CU/O-DU/O-RU architecture; RAN Intelligent Controller (RIC — non-RT and near-RT, xApps/rApps); Cloud RAN/vRAN and virtualization (NFV, Docker/Kubernetes); multi-vendor ecosystem planning/troubleshooting; O-RAN security architecture; Phase 4 milestone assessment.

Phase 5 - AI in the RAN, Days 29-35: Self-Organizing Networks (SON) and Intent-Based Networking; Python for telecom data analytics (Pandas/NumPy/Matplotlib); network programmability via REST APIs; AI for traffic prediction (ARIMA/LSTM); AI-driven energy saving (green telecom); intelligent beamforming/dynamic slice allocation via reinforcement learning; Phase 5 milestone assessment (clean automation script delivery).

Phase 6 - E2E Solutions Engineering & Capstone Project, Days 36-42: capacity engineering for special events (Cells on Wheels); Capstone Project — design/optimize an end-to-end Smart City Network featuring native 5G SA Core, virtualized O-RAN, network slicing, and AI-driven automation; iterative design reviews; pre-defense; Grand Capstone Defense before an executive evaluation panel (Technical Accuracy 30%, Practical Lab Validation 30%, Automation/AI Integration 20%, Presentation 20%).

Hands-On: Real-life practical equipment used in mobile provider fields; live 3G/4G Legacy Lab and Advanced 5G Standalone Native Cloud Lab.

8.2 5G/6G Open RAN Technologies — Code: W004

Technical Program Duration: 294 Hours

Objectives

Path to O-RAN and RAN functional splits; Split 7.2x for O-RAN; O-RAN virtualization; Non-RT and Near-RT RIC; Slicing in O-RAN.

Prerequisites

- Electronics and Communications Engineers.
- Computer Engineers.

Contents

Mobile networks survey (AMPS, GSM, UMTS, HSPA, LTE); 5G network architecture, technologies, applications; cloud/virtualization/containerization concepts.

Open RAN sections: 1. Design Goals of Open RAN. 2. Evolution to Open RAN — traditional to contemporary basestation, vRAN, distributed vs centralized RAN, horizontal disaggregation. 3. RAN Splits (Logical View) — functional split options (1, 2, 6, 8), Option 7.2x, enhanced CPRI (e-CPRI). 4. Overview of O-RAN Architecture — Open RAN vs O-RAN, control/user plane disaggregation, O-RAN entities/interfaces, control loops. 5. Virtualization Techniques for O-RAN — PNFs to VNFs, cloud-native network functions, O-Cloud deployment, DevOps CI/CD, network automation for CAPEX/OPEX reduction. 6. Detailed O-RAN Architecture — SMO, non-RT RIC, JSON/A1 policy formats, near-RT RIC (centralized/distributed). 7. O-RAN Traffic Steering Use Case — KPI data collection, roles of non-RT/near-RT RIC and E2-node. 8. Network Slicing in 5G O-RAN — S-NSSAI, NSSI, CSMF/NSMF/NSSMF, Network Slice Template, ONAP architecture and slice-instance creation. 9. Other O-RAN Use Cases — NSSI resource allocation optimization; flight-path-based dynamic UAV radio resource allocation.

9. Category: Software Engineering

9.1 Enterprise Java Developer — Code: S014

Technical Program Duration: 294 Hours (42 training days)

Overview

Transforms Computer Engineering/Science graduates into job-ready Java developers, combining foundational Java with modern enterprise technologies: Spring Boot, Java EE 7 (via RedHat Academy resources), RESTful/SOAP web services, databases, frontend integration, and deployment.

Target Audience / Prerequisites

- Computer Engineering/Science graduates or developers transitioning from other languages (C#, Python, JavaScript).
- Basic programming, OOP concepts, and basic SQL knowledge.

Contents (by Week)

Weeks 1-2 - Java Foundations & SOLID: environment/Java basics; OOP refresher; Collections Framework; SOLID principles; generics/Comparable-Comparator; lambdas and Streams API; Maven/Git/JUnit 5; exception handling and logging (SLF4J/Logback); JDBC and transactions.

Week 3 - Traditional Web Technologies: Java Servlets (lifecycle, sessions, filters); JSP & MVC pattern.

Java EE 7 & EJB (RedHat Academy): Java EE architecture, session beans (@Stateless/@Stateful/@Singleton); CDI (dependency injection, scopes, interceptors, events); JPA persistence and transactions (CMT/BMT, JTA); web services and messaging (JAX-RS, JAX-WS, MDB/JMS, @Asynchronous).

Week 4 - Spring Framework Core: DI/IoC, XML config; bean lifecycle/scopes; Java-based configuration (@Configuration, @Bean, component scanning); advanced Spring Core (profiles, layered apps).

Week 5 - Spring Boot: auto-configuration, @SpringBootApplication, embedded Tomcat; Spring Boot MVC (@RestController, validation); production-ready features (profiles, @ConfigurationProperties, logging,

@ControllerAdvice, file upload/download).

Week 6 - Spring Data JPA: JPA/Hibernate fundamentals; entity relationships (@OneToOne/@OneToMany/@ManyToMany, cascade/fetch types); Spring Data JPA repositories (query methods, JPQL, native SQL); transactions and performance (N+1 problem, DTO pattern).

Week 7 - Web Services: REST principles and best practices; RESTful API development with Swagger/OpenAPI; SOAP web services (WSDL, contract-first, Spring Boot SOAP); message queues (RabbitMQ).

Week 8 - Frontend Development: HTML5/CSS3 (semantic elements, Flexbox/Grid, responsive design); JavaScript and integration (DOM, Fetch API, Promises/async-await, Bootstrap); Linux for Java developers (fundamentals, deployment workflows).

Week 9 - Advanced Topics & Career Prep: Spring Security & JWT; testing and QA (Mockito, @SpringBootTest, MockMvc, 70%+ coverage target); Docker & microservices (Dockerfile, Docker Compose, microservices intro); final project and interview preparation.

Final Project: Full-stack application (E-Commerce, Banking, or Hospital Management System) with choice of Spring Boot or Java EE 7, RESTful APIs with documentation, SOAP integration, JPA with complex relationships, security implementation, frontend (HTML/CSS/JS/Bootstrap), exception handling/logging, 70%+ test coverage, Swagger docs, MySQL/PostgreSQL, Docker containerization, Linux deployment, and Git version control.

Job Profiles: Software Developer, Java Back-End Developer, Java Developer, Java Spring Developer, Java Enterprise Developer.

9.2 Application Development and Integration — Code: I008

Technical Program Duration: 294 Hours

Overview

Focuses on Service-Oriented Architecture (SOA) and effective information management. After a Fundamentals module, trainees branch into either Enterprise Content Management (ECM) or Business Process Management (BPM).

Prerequisites

- Fair knowledge of APIs and integration.
- Software development and enterprise-solution experience.
- Very good knowledge of SDLC.
- Fair knowledge of web development.
- Good knowledge of Java development.

Contents

Fundamentals (fast recap): Active Directory/LDAP (8 hr); OOP and data-structure recap (16 hr); database recap — schemas, tables, views, relations, SQL (16 hr); Information Management overview — ECM vs BPM concepts and use cases (8 hr).

Enterprise Content Management (ECM) Intake: - *IBM Technology:* FileNet P8 Platform Administration & Configuration (40 hr) — architecture, ACCE UI, object stores, taxonomy, security, search/bulk actions; FileNet P8 Development & Troubleshooting (40 hr) — APIs, subscriptions, sweep tasks, standalone tools, troubleshooting; IBM Content Navigator (40 hr) — architecture, repositories/desktops, entry/search templates, customization, External Data Service, custom actions; IBM DataCap (40 hr) — content capturing, Server/Application Manager, Rule Runner, Web Service, DataCap Studio, application templates, rulesets/workflows. - *OpenText Technology:* OpenText Content Server Administration & Implementation (40 hr) — architecture, Smart/Classic UI, taxonomy, search, categories, classifications, users/permissions/ACLs; OpenText APIs Developing (40 hr) — workflows (definitions vs instances), monitoring, assignments, LiveReports/WebReports, REST APIs.

Business Process Management (BPM) Intake: - IBM BPM Level 1 (80 hr) — architecture/components, process

applications, business objects/data model, gateways, teams, four-eyes policy, coach views, human client services. - IBM BPM Level 2 (80 hr) — DB integration, nested human services, coach-view events, team distribution, exception handling, theme UI, EPV/ENV/localization, team filter service, loops, sub-processes/linked processes. - IBM BPM Level 3 (80 hr) — web services, mail/LDAP integration, ECM integration (document attachments), BPM REST APIs, Under Cover Agents, tracking groups, instance-view customization, snapshot deployment, process branching, toolkits.

9.3 Big Data Engineering — Code: I009

Technical Program Duration: 294 Hours

Overview

Covers Linux essentials for Hadoop platforms, Python programming (Pandas, Scikit-learn), SQL, and the big-data ecosystem (HDFS, HBase, Hive, MapReduce, YARN, Spark, Flink, Flume, Kafka, Elasticsearch, Zookeeper), enabling trainees to store/manage/analyze data at scale and plan a big-data strategy.

Target Audience / Prerequisites

- Communications/Electronics or Computer Engineering, Information/Computer, or Computer Science graduates.
- Network Essentials.

Contents

Linux Essentials (28 hr); Python for Data Analysis (35 hr); SQL Essentials (21 hr, incl. PostgreSQL); Huawei HCIP Big Data H13-723 (100 hr) — big-data application development, HDFS, MapReduce/YARN, Zookeeper, HBase, Elasticsearch, Spark2x/SparkSQL, Flink, Sqoop/Flume/NiFi/Airflow, data warehousing (DWS, Hive), Kafka; Machine Learning and Data Analysis (69 hr) — data-mining lifecycle (business/data understanding, preparation, modeling, evaluation, deployment), classification/segmentation/ML models, supervised/unsupervised learning, neural networks, Power BI/R Markdown/Tableau; Cloud Computing — AWS fundamentals (35 hr) — IAM, EC2, storage/database services, VPC, CloudFormation, management/security tools.

Job Profiles: Data Scientist/Big Data Engineer, Data Analyst, Data Management and Mining, Machine Learning Engineer.

9.4 AI in Data Analysis — Code: S015

Technical Program Duration: 294 Hours

Overview

Prepares participants for modern data analytics, BI, and AI-driven decision-making roles, combining data analysis, cloud analytics, data engineering, business intelligence, and generative AI. Begins with Linux/automation fundamentals, then AWS cloud analytics, Python programming, statistical/EDA methods, SQL/data engineering, Generative AI & LLMs for analytics, and big-data/BI tools (Spark/PySpark, Power BI).

Certification References

IBM Python for Data Science AI & Development; Google Data Analytics Professional Certificate; IBM Data Engineering Professional Certificate; IBM Databases and SQL for Data Science; IBM Generative AI Engineering Certificate; IBM Machine Learning with Apache Spark; Databricks Fundamentals; IBM Data Analyst Professional Certificate; Microsoft Power BI Learning Path; AWS Cloud Practitioner; AWS Data Engineering Associate.

Target Audience / Prerequisites

- Communications/Electronics or Computer Engineering, Information/Computer, or Computer Science graduates.

- Network Essentials.

Contents

Module 1 – Linux & Automation (21 hr); Module 2 – AWS Cloud Practitioner (35 hr); Module 3 – Python for Data Analysis (35 hr); Module 4 – Git and GitHub (7 hr); Module 5 – Data Preparation & Advanced Programming Libraries (35 hr, incl. NumPy/Pandas, Excel/Power Query); Module 6 – Mathematical Foundations, Statistics & EDA (14 hr, incl. hypothesis testing, A/B testing, KPI analysis); Module 7 – Network Data Ingestion & Data Engineering Pipelines (42 hr, incl. REST APIs, web scraping, advanced SQL, data warehousing/ETL, Airflow/NiFi, data governance); Module 8 – Generative AI & LLMs for Data Insights (42 hr, incl. prompt engineering, AI-assisted SQL/Python/cleaning, RAG, AI agents, automated reporting, responsible AI); Module 9 – Big Data Analytics, Cloud Analytics & BI (35 hr, incl. Spark/PySpark, Power BI/DAX/star schema/dashboards); Module 10 – Data Analytics Capstone Project (28 hr).

Job Profiles: AI Data Analyst, Business Intelligence Analyst, Data Analyst, Product Data Analyst, Junior Data Engineer, Reporting Analyst.

9.5 DataOps Engineering — Code: I016

Technical Program Duration: 294 Hours

Objectives

Equips trainees for careers in Data Engineering, ETL Development, Data Pipeline Engineering, Cloud Data Platforms, Data Automation, and Big Data Systems — designing, building, automating, and managing scalable data pipelines and modern data infrastructure.

Who May Apply

Graduates of Engineering/Science (Computer Science, Statistics, Mathematics), Computing and Informatics, Computing and AI, or Computing and Data Science.

Contents

Module 1 – Python for Data Engineering (fundamentals, OOP, file handling, APIs, JSON/XML, Pandas/NumPy, logging); Module 2 – SQL & Advanced Databases (joins, subqueries, window functions, stored procedures, query optimization, relational vs NoSQL); Module 3 – Data Warehousing & Modeling (star/snowflake schema, fact/dimension tables, OLTP vs OLAP, data lakes/lakehouse); Module 4 – ETL & Data Pipelines (ETL/ELT, batch processing, validation, incremental loading, cleaning pipelines); Module 5 – Big Data Fundamentals (ecosystem, Hadoop, HDFS, distributed systems, scalability); Module 6 – Apache Spark & Distributed Processing (PySpark, RDDs, DataFrames, Spark SQL, optimization); Module 7 – Data Automation & Workflow Orchestration (scheduling, event-driven pipelines, monitoring/alerts, CI/CD for data, data quality automation); Module 8 – Cloud Data Engineering (cloud storage, data factory, managed Spark, cloud pipelines/automation); Module 9 – Streaming Data Engineering (real-time pipelines, stream processing, Kafka basics); Module 10 – Capstone Projects.

Job Profiles: Data Engineer, ETL Developer, Big Data Engineer, Cloud Data Engineer, DataOps Engineer.

10. Frequently Asked Questions (FAQ)

1. Who is eligible to apply for the scholarship? Anyone who has not graduated more than five years ago.

2. Does studying require attending the institute's premises? Yes, the initiative is available through the attendance system at the institute's locations.

3. Is there practical training or is the study purely theoretical? The study includes both theoretical and practical aspects; practical training is conducted based on the training-unit content using the latest methods and specialized programs.

4. Are employees allowed to join the scholarship? Full commitment to attending lectures daily throughout the training unit is required to be eligible.

5. What is the duration of the training unit? 3 months at the institute's premises, plus 1 month of practical training at one of the companies supporting the selected training unit.

6. If I previously benefited from the institute's scholarships (6-month or 3-month), can I apply for the new training units? You can only benefit from the "HireReady" and "Digital Egypt Youth" initiatives once.

7. Do I need a computer to join the scholarship? Preferable to have a personal laptop; if not possible, NTI provides a computer throughout the training period at the premises.

8. When will I be contacted for the interview? Once the application period closes, you will be contacted to schedule acceptance tests, which require your presence at the premises on the specified date.

9. When will I be notified of the interview results? Within two weeks after the interview. If accepted and the required number of participants is complete, you'll be contacted to confirm paperwork and readiness to participate.

10. What should I do if I entered incorrect information in the registration form? Email D4m@nti.sci.eg with your full name and national ID number for follow-up.

11. How can I change my preferences? Email D4m@nti.sci.eg requesting deletion of all preferences, then enter new preferences using the same account. Modifications are treated as a new registration starting from the modification date.

12. When is an applicant considered accepted for the scholarship? Three stages: (1) meeting all training-unit requirements listed on the registration page; (2) completing the personal interview; (3) submitting all required documents.

13. How can I withdraw from the scholarship? Send a withdrawal request at least 48 hours before training starts, including full name, national ID number, training unit applied for, and registration number, to D4m@nti.sci.eg.

14. Can non-Egyptian nationals apply for the scholarship? The scholarship is only available for Egyptians.

15. Details about the 4-month scholarship (attendance, duration, online/offline)? 5-day work week, 7 working hours/day, morning sessions, attendance required at the institute's premises throughout. NTI provides free transportation from anywhere in Cairo to the premises during the initiative.

16. What is the deadline for submitting the certificate? Before the start of the acceptance-test phase.

17. Will financial incentives be provided during training? Yes, a monthly allowance is provided throughout the 3-month training period at the premises.

18. Does the scholarship require a specific GPA? No specific GPA is required — only selecting an eligible specialty and passing the acceptance tests.

11. Contact Information

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- **Smart Village:** Tel: (+202) 35342423 / 35342424 / 35342452 — Fax: (+202) 35342483
- **Preferences/registration corrections:** D4m@nti.sci.eg